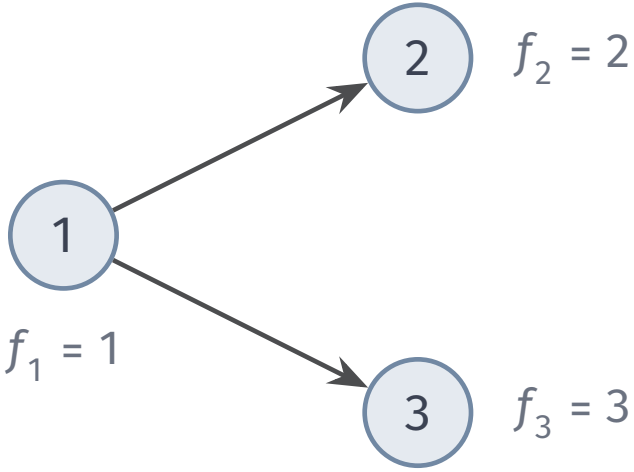

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\mathbf{f} = (1, 2, 3)^T$

Af

$(5, 0, 0)^T$

$\mathbf{A}^T\mathbf{f}$

$(0, 1, 1)^T$